

WAUSAU, WI, USA

WMO: 726463

Lat: 44.929N

Lon: 89.628W

Elev: 366

StdP: 97.01

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14897

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -24.5	(c) -21.9	(d) -29.5	(e) 0.3	(f) -23.8	(g) -26.6	(h) 0.3	(i) -21.3	(j) 9.9	(k) -12.3	(l) 8.9	(m) -7.7	(n) 2.8	(o) 300	(p) 0.574

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	30.9	21.9	29.1	20.8	27.7	19.9	23.5	28.4	22.5	27.1	21.5	25.8	4.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.1	17.5	26.1	21.0	16.3	25.2	19.9	15.3	24.2	72.0	28.3	67.8	27.1	64.2	25.7	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.3	(b) 8.2	(c) 7.4	(e) -28.3	(f) 33.5	(g) 2.8	(h) 2.1	(i) -30.3	(j) 35.1	(k) -32.0	(l) 36.3	(m) -33.5	(n) 37.5	(o) -35.6	(p) 39.0		
(a) 9.3	(b) 8.2	(c) 7.4	(e) -28.5	(f) 25.2	(g) 2.7	(h) 1.4	(i) -30.4	(j) 26.2	(k) -32.0	(l) 27.0	(m) -33.6	(n) 27.8	(o) -35.5	(p) 28.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.9	-9.2	-7.2	-0.7	6.5	13.5	18.9	21.2	20.0	15.7	8.5	0.9	-6.0		
	DBStd	11.88	6.82	6.78	6.58	5.36	4.70	3.83	3.12	3.05	4.44	4.81	5.41	6.28		
	HDD10.0	2429	597	483	339	129	19	0	0	0	6	87	274	496		
	HDD18.3	4431	855	716	592	356	163	39	9	18	102	306	522	754		
	CDD10.0	1302	0	0	6	24	127	267	347	309	178	42	3	0		
	CDD18.3	261	0	0	0	1	13	55	98	68	24	2	0	0		
	CDH23.3	2221	0	0	1	14	147	486	856	510	194	13	0	0		
	CDH26.7	572	0	0	0	2	34	131	254	110	39	1	0	0		
(14) Wind	WSAvg	3.4	3.5	3.5	3.5	3.8	3.6	3.2	3.0	2.9	3.1	3.5	3.6	3.4		
(15) Precipitation	PrecAvg	804	26	23	46	72	104	100	96	106	100	58	47	33		
	PrecMax	1059	79	57	118	127	194	239	236	235	221	142	111	77		
	PrecMin	544	2	1	6	22	35	13	37	22	16	9	0	10		
	PrecStd	143	20	16	32	34	44	54	51	49	53	31	29	19		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.3	9.6	21.0	25.7	30.0	32.6	34.0	31.5	30.2	25.3	17.6	10.3		
		MCWB	3.3	5.8	14.6	15.8	19.4	22.5	24.2	22.7	21.6	18.1	11.8	8.3		
	2%	DB	3.2	6.2	15.7	21.6	27.2	29.8	31.3	29.3	27.7	22.1	13.8	5.9		
		MCWB	1.4	3.3	10.4	12.6	17.7	21.3	22.4	21.2	20.1	16.4	9.9	3.5		
	5%	DB	1.4	3.7	11.5	18.5	24.7	28.0	29.3	27.7	25.7	19.2	11.2	3.5		
		MCWB	0.3	1.6	7.5	10.6	16.0	19.7	21.2	20.2	18.8	14.1	7.7	1.9		
	10%	DB	0.1	1.9	8.0	15.8	22.1	26.2	27.7	26.3	23.5	16.6	8.8	1.7		
		MCWB	-0.8	0.1	4.5	8.8	14.7	18.7	20.1	19.4	17.8	12.5	6.1	0.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	7.0	14.9	16.8	21.5	24.0	25.6	24.6	22.8	20.4	13.3	8.9		
		MCDB	5.5	9.2	19.9	23.0	27.5	29.6	31.7	28.9	27.6	23.2	15.9	10.1		
	2%	WB	1.6	3.6	11.5	14.0	19.3	22.6	23.9	22.9	21.4	17.1	10.6	4.1		
		MCDB	3.0	5.8	14.4	20.0	24.5	28.3	28.9	27.0	25.5	20.0	13.1	5.3		
	5%	WB	0.5	1.7	7.7	11.5	17.6	21.3	22.7	21.8	20.2	15.2	8.0	2.0		
		MCDB	1.4	3.6	11.3	16.7	22.5	26.3	27.2	25.5	24.1	18.7	10.5	3.4		
	10%	WB	-0.9	0.1	4.8	9.6	15.8	20.1	21.6	20.8	18.9	12.9	6.2	0.6		
		MCDB	0.2	1.7	7.9	15.3	20.6	24.3	26.0	24.5	22.2	16.1	8.7	1.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	9.0	9.7	11.0	11.6	11.0	11.1	10.6	11.0	9.5	7.7	7.2		
		MCDBR	8.1	9.6	13.0	16.2	14.5	13.0	12.4	11.9	13.1	13.4	11.6	8.9		
	5% WB	MCWBR	7.0	7.3	8.3	8.9	6.8	5.8	5.3	5.3	6.4	8.0	8.2	7.3		
		MCWBR	7.6	8.9	12.2	14.1	12.1	11.5	11.2	10.1	10.7	11.8	10.7	7.8		
(40) Clear-Sky Solar Irradiance	taub		0.247	0.278	0.317	0.346	0.381	0.381	0.374	0.376	0.350	0.315	0.286	0.255		
	taud		2.393	2.300	2.357	2.377	2.320	2.345	2.376	2.372	2.447	2.532	2.523	2.447		
	Ebn at Noon		889	911	918	915	888	887	888	874	872	865	837	843		
	Edh at Noon		72	96	106	114	124	122	117	113	97	77	65	62		
(44) All-Sky Solar Radiation	RadAvg		1.64	2.75	3.91	4.56	5.40	5.89	6.06	5.16	4.07	2.55	1.58	1.18		
	RadStd		0.18	0.26	0.41	0.46	0.43	0.45	0.31	0.34	0.29	0.27	0.17	0.13		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page